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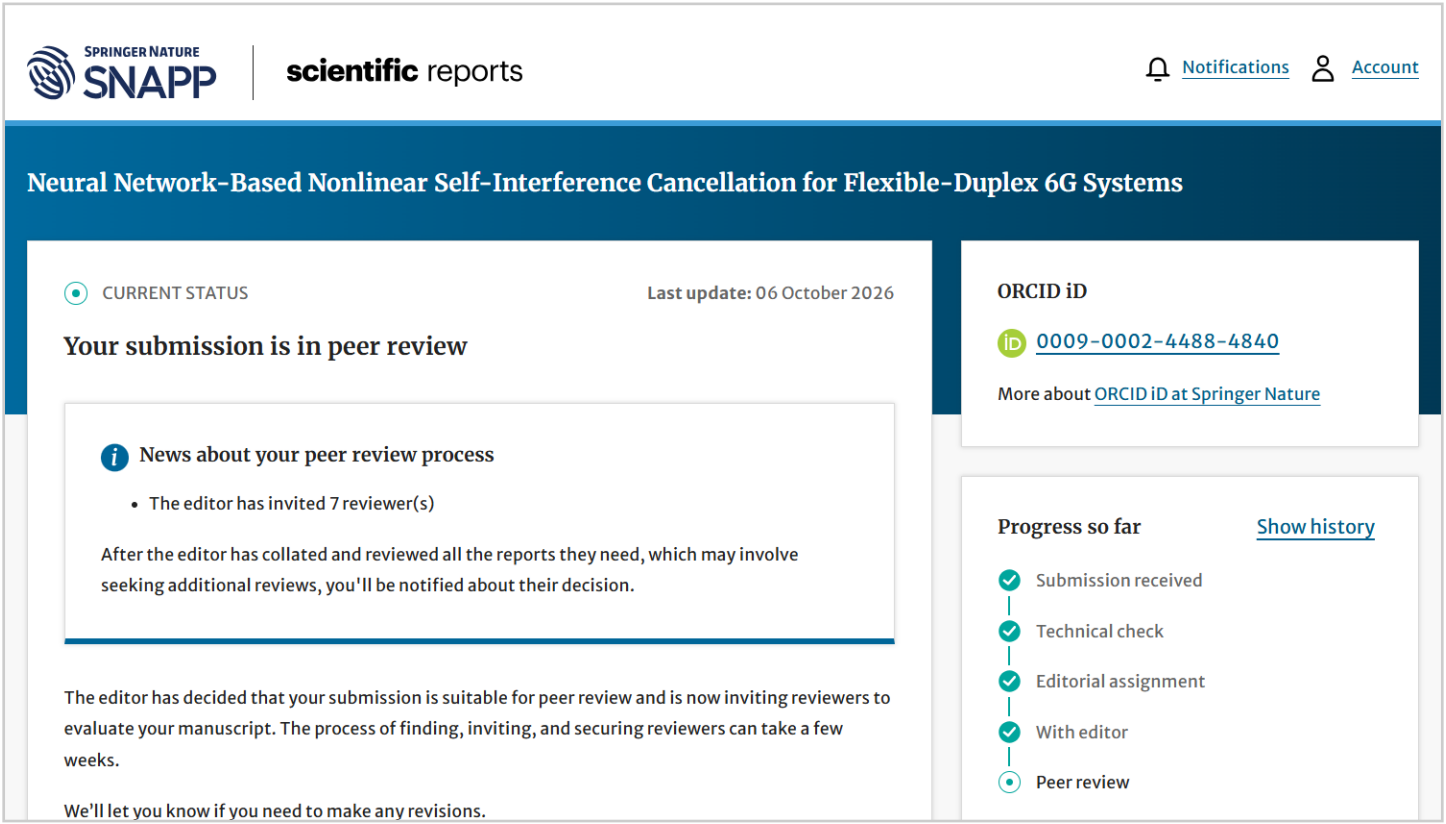
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Neural Network-Based Nonlinear Self-Interference Cancellation for Flexible-Duplex 6G Systems

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Abstract

Future 6G wireless networks will encounter highly asymmetric traffic, with downlink (DL) transmissions dominating uplink (UL). Conventional duplexing methods lack the flexibility to efficiently support such dynamic patterns. Subband full duplex (SBFD) and 6G flexible-duplex address this challenge by enabling simultaneous UL/DL transmissions with dynamic subband allocation and partial or complete overlap. However, these approaches introduce leakage from the transmitter to the receiver, leading to severe self-interference (SI) at the receiver, which is further exacerbated by nonlinear distortion from the transmitter power amplifier (PA). In addition to short-term memory, modern GaN HEMT-based PAs produce long-term memory effects due to thermal and trapping dynamics, causing the leaked signal to contain both nonlinearity and memory-induced distortion. This paper proposes a neural network (NN)-based SI cancellation framework for 6G flexible-duplex systems. The NN accurately models transmitter's PA nonlinearities, short and long-term memory effects, and receiver low-noise amplifier (LNA) distortion. An adaptive modeled signal provides analog-domain suppression before the LNA, while digital cancellation eliminates residual SI. The proposed approach achieves over 100 dB SI suppression, enabling efficient and practical 6G flexible-duplex operation.

Keywords: 6G Communication, artificial intelligence, analog, digital cancellation, flexible-duplex, GaN HEMTs, machine learning, memory effects, neural network, nonlinearity, power amplifier, self-interference, wireless communication

1 Introduction

Future mobile/wireless systems (6G) are expected to experience highly asymmetric traffic patterns, with downlink (DL) transmissions dominating uplink (UL) due to the increasing demand for high-rate applications such as video streaming, extended reality, and holographic communication [1]. Conventional duplexing schemes, such as frequency-division duplex (FDD) and time-division duplex (TDD), offer limited adaptability in handling such imbalances. To address this, subband full duplex (SBFD) has been introduced as a promising technique [2]. In SBFD, the base station (BS) allocates separate UL and DL subbands within a common carrier according to instantaneous traffic requirements. The subbands remain non-overlapping, as illustrated in Fig. 1(a), enabling simultaneous transmissions while maintaining manageable interference levels [3].

The SBFD approach can be generalized into what is referred to as 6G flexible-duplex, envisioned as a candidate duplexing paradigm for 6G systems [4]. 6G Flexible-duplex extends SBFD by permitting partial or complete overlap between UL and DL frequency resources, as shown in Fig. 1(c). Since self-interference (SI) cancellation is already necessary in SBFD, the additional complexity of enabling overlap is minimal, while the potential spectral efficiency gain is significant [5]. Furthermore, consistent with the principles established in 5G New Radio (NR), flexible-duplex allows dynamic configuration of DL component carriers (CCs), either contiguously or non-contiguously, across the available spectrum [6].

While SBFD and 6G flexible-duplex are motivated by the need to accommodate asymmetric traffic, the broader concept of in-band full duplex (IBFD) provides the theoretical upper bound on spectrum utilization [5]. In IBFD, a transceiver simultaneously transmits and receives on the exact same frequency resources, which in principle can double spectral efficiency compared with FDD and TDD [7, 8]. The critical obstacle to realizing IBFD is strong SI, which can exceed the desired receiver signal strength by over 100 dB [8]. Over the past decade, extensive research has explored SI suppression techniques spanning the RF/analog, and digital baseband domains, moving IBFD closer to practical feasibility [9–11].

In both SBFD and 6G flexible-duplex systems, the transmit signals (DL), whether occupying contiguous or non-contiguous component carriers, experience nonlinear distortion after passing through the transmitter's power amplifier (PA). In practical implementations of compact and low-cost flexible-duplex transceivers, the transmitter and receiver commonly share a single antenna via a circulator or employ two closely spaced antennas. Even with an excellent circulator achieving an isolation of 30 dB between the transmitter and receiver, these PA's nonlinearities leak from transmitter DL and spreads into the frequency components of the receiver UL, as shown as in Fig. 1(b) and Fig. 1(d). This leakage of transmitter's distortion to receiver of same base station is known as SI. The SI from transmitter's PA to receiver's low-noise amplifier (LNA) generates IMDs that can desensitize the receiver.

Modern wireless transmitters typically employ GaN-based high electron mobility transistor (HEMT) PAs, which inherently exhibit long-term memory effects arising from thermal dynamics and trapping phenomena [12–14]. Consequently, the leakage

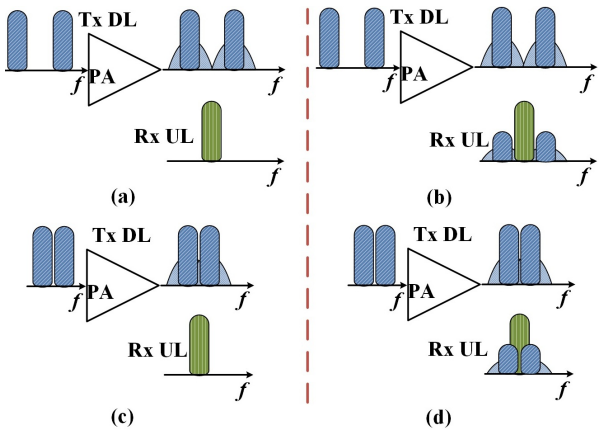


Fig. 1 Frequency-domain resource allocation for asymmetric (a) SBFD before Tx leakage, (b) SBFD after Tx leakage, (c) 6G flexible-duplex before Tx leakage, and (d) 6G flexible-duplex after Tx leakage, showing transmitter UL and receiver DL at base station.

from the transmitter is not limited to static nonlinearities but also includes memory-induced distortions. The mitigation of PA-induced nonlinearities has been investigated in [15–20]; however, these works focus solely on analog/RF, domain cancellation at the transmitter and do not account for SI effects at the duplexer.

To address this challenge, this paper presents a novel architecture and neural network (NN) model for SI cancellation in 6G flexible-duplex systems. The proposed NN model is designed to accurately capture the nonlinear characteristics of the GaN PA, the associated short and long-term memory effects, and the nonlinear distortion introduced by the receiver low-noise amplifier (LNA). For analog-domain suppression, the NN generates a modeled signal that is injected prior to the LNA, thereby mitigating front-end analog interference. In addition, to achieve SI suppression exceeding 100 dB, the same NN model is adaptively employed to map the transmitter baseband input to the receiver LNA baseband output. The resulting modeled signal is then utilized in the digital domain to cancel the residual SI, which is mainly due to the residual leakage between the transmitter and receiver.

The following section briefly analyzes the transmitter leakage, receiver nonlinearity and the proposed NN based architecture for distortion cancellation in the 6G Flexible-Duplex communication. The RF measurement setup along with the corresponding experimental results are detailed in Section 3, and the conclusions are presented in Section 4.

2 6G Flexible-Duplex Communication

2.1 Transmitter Leakage

In transmitter, the output of PA comprises of nonlinear distortion, an amplified input signal, long and short memory effects affecting the integrity of the signal. The output of PA is mathematically expressed as

$$y_{PA}(n) = f(u_{Tx}(n), Th(n), Tr(n)) \tag{1}$$

where $u_{Tx}(n)$ and $y_{PA}(n)$ are the discrete-time baseband input and baseband PA's output signals at the transmitter, sampled at the frequency f_s. f is the nonlinear function with memory effects. $Th(n)$ is the discrete-time modeled junction/channel temperature due to the thermal effects, which results in long-term memory [12]. It can be mathematically expressed as

$$Th(n) = (1 - \phi_{th})Th(n-1) + \phi_{th}|u_{Tx}(n)|^2 \tag{2}$$

where ϕ_{th} is specified as

$$\phi_{Th} = 1 - e^{-\frac{T_s}{\tau_{th}}} \tag{3}$$

where T_s is the sampling time step and τ_{th} is the thermal time constant of the amplifier.

The two branch RC circuit is used for modeling of trapping potential Tr(n) to reduce the trapping effects [13, 14], and it is formulated as:

$$Tr(n) = \begin{cases} (1-\phi_c)Tr(n-1) + \phi_c|u_{Tx}(n)|, & |u_{Tx}(n)| > Tr(n-1) \\ (1-\phi_e)Tr(n-1) + \phi_e|u_{Tx}(n)|, & |u_{Tx}(n)| \leq Tr(n-1) \end{cases} \tag{4}$$

where

$$\begin{aligned} \phi_c &= 1 - e^{-\frac{T_s}{\tau_c}} \\ \phi_e &= 1 - e^{-\frac{T_s}{\tau_e}} \end{aligned} \tag{5}$$

where τ_e and τ_c are the emission and capture time constants of the PA, and T_s is the sampling time step.

The circulator is used for isolation between transmitter and receiver signal. However, due to limited isolation of the circulator, the transmitter signal is leaked to the receiver. This leaked transmitter signal will result in SI at receiver of 6G flexible-duplex. The transmitter leakage signal is formulated as

$$y_{TxLe}(n) = \sum_{l=1}^{L} h_l y_{PA}(n - \tau_l) \tag{6}$$

where h_l comprise the direct transmitter–receiver leakage, reflections due to antenna mismatch, and additional reflections originating from the surrounding environment. $y_{TxLe}(n)$ is the transmitter leakage signal.

2.2 Receiver's LNA Nonlinearity

In receiver's path of 6G flexible-duplex, the receiver signal would combine with the transmitter leakage. Thus the received signal with SI is mathematically expressed as

$$y_{Rx}(n) = u_{Rx}(n) + y_{TxLe}(n) \tag{7}$$

where $u_{Rx}(n)$ is the incoming baseband receiver signal UL, and y_{Rx} is the baseband received signal with SI. After this, the received signal would pass through the LNA. The output of the LNA is formulated as

$$y_{LNA}(n) = h\left(u_{Rx}(n)\right) + h\left(y_{TxLe}(n)\right) \tag{8}$$

where h is the non-linear function associated with LNA. The nonlinearity of LNA is less severe than the nonlinearity of transmitter's PA.

2.3 Essential need for 6G Flexible-Duplex

In 6G flexible-duplex, the non-desired signal must be mitigated out at the receiver, i.e. the leaked transmitter signal must be cancelled from the 6G flexible-duplex. Thus the output of LNA can also be written as

$$y_{LNA}(n) = h\left(u_{Rx}(n)\right) + h\left(y_{TxLe}(n)\right) = h\left(u_{Rx}(n)\right) + y_{SI}(n) \tag{9}$$

where

$$y_{SI}(n) = h\left(y_{TxLe}(n)\right) \tag{10}$$

The $y_{SI}(n)$ signal is characterized as a function of the transmitter PA's nonlinearity, including its short and long-term memory effects, in conjunction with the receiver LNA's nonlinearity. This non-desired signal should be minimized.

The $y_{SI(n)}$ signal is highly complex to model; therefore, a NN-based model is employed in the subsection 2.5 characterize it.

2.4 Proposed Architecture for Distortion Cancellation in 6G Flexible-Duplex

Fig. 2 shows the architecture for the cancellation of distortion in 6G Flexible Duplex. In the first stage, the proposed NN-based model is employed to characterize the relationship between the transmitter baseband input $u_{Tx}(n)$, $Th(n)$, $Tr(n)$, and the receiver LNA baseband output $y_{LNA}(n)$. The modeled output with adjusted phase is then transmitted to the receiver through an additional transmitter (DAC_2) and combined with the received signal prior to the LNA using an RF power combiner. Thereby suppressing transmitter leakage in the analog/RF domain i.e. analog cancellation. In the subsequent stage, the LNA output is re-captured, where the residual SI is further reduced compared to the previous iteration. The proposed adaptive NN-based model is again utilized to map the transmitter baseband input to the receiver LNA baseband output, and the resulting modeled signal is employed to cancel the residual SI in the digital domain i.e. digital cancellation.

2.5 Proposed NN-Based Adaptive Model for Distortion Cancellation

Let us consider $I_{Tx}(n)$ and $Q_{Tx}(n)$ are the real and imaginary parts of the transmitter baseband contiguous or non-contiguous input signal $u_{Tx}(n)$. The input vector $\mathbf{X_1}$ comprises of real, imaginary, magnitude and delays of the transmitter's input signal.

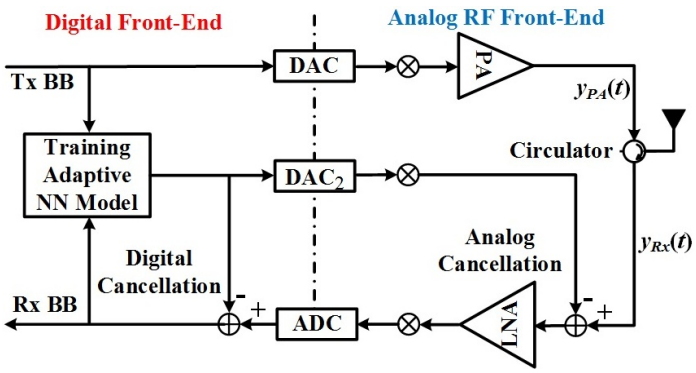


Fig. 2 Proposed Architecture for Distortion Cancellation in 6G Flexible Duplex.

$$\mathbf{X_1} = \left[I_{Tx}(n)\ Q_{Tx}(n)\ |u_{Tx}(n)|\ Delays\right] \tag{11}$$

First the features $Th(n)$ and $Tr(n)$ are extracted using ϕ_{th}, ϕ_e and $|u_{Tx}(n)|$ as defined in Eq.(2) and Eq.(4), respectively. The thermal time constant τ_{th} is modeled through a learnable neuron weight (W_{th}). In practice, τ_{th} typically lies in the millisecond range [13]. Thus ϕ_{th} is formulated as

$$\phi_{th} = 1 - e^{-\frac{T_s}{\tau_{th}}} = 1 - e^{-\frac{T_s}{W_{th}\times 10^{-3}}} \tag{12}$$

Similarly, the emission time constant τ_e is modeled using a learnable neuron weight (W_e). Generally, τ_e lies in the microsecond range [13]. Hence ϕ_e is formulated as

$$\phi_e = 1 - e^{-\frac{T_s}{\tau_e}} = 1 - e^{-\frac{T_s}{W_e\times 10^{-6}}} \tag{13}$$

Both the $Th(n)$ and $Tr(n)$ are parameterized as functions of neuron weights, enabling them to be adaptively optimized during network training. Since these variables depend on multiple physical factors such as temperature and operating conditions, the proposed neural network framework facilitates effective data-driven learning of the underlying memory effects.

Fig. 3 illustrates the feature extraction stage in the input layer of the proposed NN architecture, where recursively updated state variables $Th(n)$ and $Tr(n)$ are computed to capture the PA's memory effects. The function g_1 in Fig. 3 corresponds to Eq.(2) and Eq.(12), while function g_2 corresponds to Eq.(4) and Eq.(13). These nonlinear features are concatenated with signal vector $\mathbf{X_1}$ and passed through multiple fully connected layers. The resulting concatenated input vector is expressed as

$$\mathbf{X_2} = \left[\mathbf{X_1}\ \mathrm{Th}(n)\ \mathrm{Tr}(n)\right] \tag{14}$$

Fig. 3 also illustrates the overall NN architecture, which employs two hidden layers to maintain low computational complexity. Each hidden layer uses the hyperbolic tangent (Tanh) f activation function, while the adaptive moment estimation (ADAM)

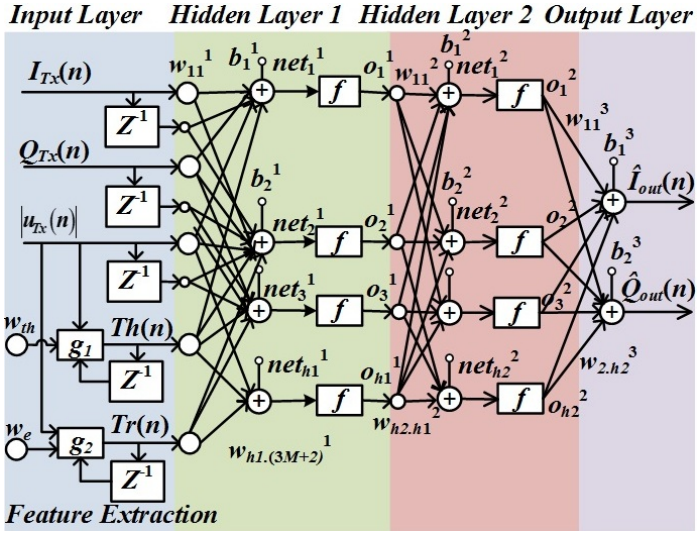


Fig. 3 Proposed NN Architecture for Distortion Cancellation in 6G Flexible Duplex.

optimization algorithm is utilized to determine the network's weights and biases. The output from each layer serves as the input to the subsequent layer, ensuring sequential feature extraction. Finally, the outputs of the second hidden layer are linearly combined at the output layer to generate the network response.

The input data $\mathbf{X_2}$ is propagated through the neural network from the input layer toward the output layer. During training, the real $I_{LNA}(n)$ and imaginary $Q_{LNA}(n)$ parts of the baseband LNA output $y_{LNA}(n)$ are used as the desired NN outputs. The network generates estimated in-phase $\hat{I}_{out}(n)$ and quadrature $\hat{Q}_{out}(n)$ signal components. The difference between the predicted outputs and the corresponding measured values ($I_{LNA}(n)$, $Q_{LNA}(n)$) is quantified using an appropriate loss function. ADAM computes the gradient of the loss function and maintains exponentially weighted estimates of both the first-order moment (mean of gradients) and the second-order moment (uncentered variance of gradients). These moment estimates are subsequently utilized to iteratively update the weights and biases. The optimization process is carried out via backpropagation, during which the error is propagated from the output layer toward the input layer, enabling systematic adjustment of weights and biases across all network layers.

The complex valued NN output is given by $\hat{y}_{out}(n)= \hat{I}_{out}(n)+\mathrm{j}\hat{Q}_{out}(n)$. Consequently, the phase-adjusted NN-modeled output, $y_{nn}(n)=-\hat{y}_{out}(n)$, is transmitted via an additional transmitter (DAC_2) to the receiver for front-end analog SI cancellation.

In the subsequent stage, the LNA output is recaptured, where the residual SI is further reduced compared to the previous iteration. The proposed NN model is again adaptively trained by $I_{Tx}(n)$, $Q_{Tx}(n)$, $Th(n)$, $Tr(n)$, and recaptured LNA baseband output. The resulting modeled signal is then utilized in the digital domain to cancel the residual SI.

The proposed NN requires $2+(3M+2)\times h_1+h_2\times h_1+2h_2$ weights and h_1+h_2+2 biases, where M denotes the memory depth, h_1 and h_2 represent the number of neurons

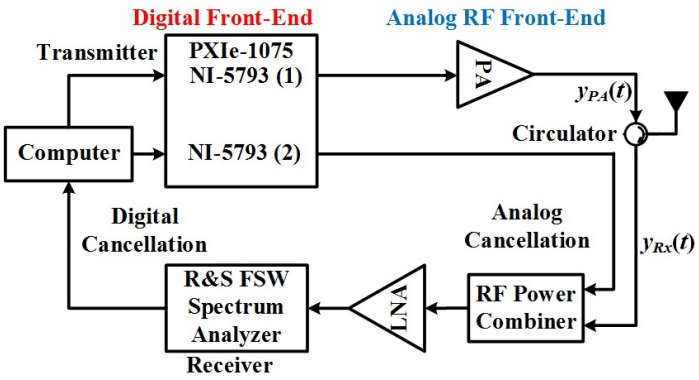


Fig. 4 RF Measurement Setup.

in the first and second hidden layers, respectively. Consequently, the overall coefficients count (weights and biases) is given by $(3M + 2) \times h_1 + h_2 \times h_1 + 2h_2 + h_1 + h_2 + 4$.

In terms of computational complexity, the hyperbolic tangent activation, real–real multiplication, real–real addition, and delay operations require 13, 1, 1, and 0 floating-point operations (FLOPs), respectively. Accordingly, the total number of FLOPs per sample is calculated as $2((3M + 2) \times h_1 + h_2 \times h_1 + 2h_2) + 13h_1 + 13h_2 + 2$.

3 RF Measurement and Results

3.1 Measurement Setup

Fig. 4 illustrates the experimental setup used for transmitter and receiver. The transmitter section employs a National Instruments (NI) PXIe-1075 chassis integrated with two RF transmitter modules (NI-5793). Both RF modules are synchronized using a 10 MHz reference clock from the PXIe-1075. The two categories of 5G-NR waveforms-specifically, continuous and non-contiguous NR-FDD signals are employed, each having bandwidth of 100 MHz and a peak-to-average power ratio (PAPR) of 8.5 dB. The DACs of the NI-5793 modules perform 2 times interpolation, resulting in a sampling rate of 491.52 Msps. The primary RF transmitter module, NI-5793 (1), upconverts the 100 MHz baseband 5G-NR signal to an RF carrier frequency of 2.14 GHz, which is then applied to the device under test (DUT). The DUT includes a wideband driver PA (Mini-Circuits ZHL-42W) and a main GaN HEMT PA (CGH40006). The driver PA operates from 10–4200 MHz with 1 dB compression at 28 dBm output and a 15 V bias, while the main PA delivers 6 W saturation power across 0.5–3 GHz with a 28 V bias.

The output of the main PA is connected through a 40 dB circulator for isolation, followed by a low-noise amplifier (LNA). The amplified signal is then captured and downconverted to baseband using an R&S FSW spectrum analyzer, which is synchronized to the same 10 MHz reference clock as the PXIe-1075. In the second iteration, the additional transmitter NI-5793 (2) generates the modeled output signal, which is combined with the received signal prior to the LNA using an RF power combiner for interference cancellation.

3.2 Measurement Results for Contiguous Signal in Flexible-Duplex

Table 1 presents the modeling results of several advanced nonlinear cancellation approaches, including those proposed by Liu [9], Kiayani [10], and Lampu [4], alongside the performance of the proposed canceller when applied to a 100 MHz contiguous 5G-NR waveform. The evaluation metrics used are normalized mean square error (NMSE) and adjacent channel error power ratio (ACEPR) [19], where the error corresponds to the difference between the modeled output and the measured LNA output. ACEPR-Left quantifies the ratio between the error's power in the left adjacent channel and the transmitted leakage signal's power in the left main channel. Likewise, ACEPR-Right reflects the same ratio for the right adjacent channel. For performance evaluation, the proposed NN model is configured with M=4, h_1=9, and h_2=8. Experimental results indicate that the proposed cancellation model achieves a lower NMSE with less computational complexity compared to the existing nonlinear cancellers for the 100 MHz contiguous 5G-NR signal. Moreover, the proposed canceller provides an enhancement greater than 6 dBc in both ACEPR-Left and ACEPR-Right, demonstrating better modeling performance relative to other referenced approaches.

Table 1 Modeling Results of Nonlinear Canceller Models for 100 MHz Contiguous 5G-NR Signal

Metric	Canceller			
	Canceller-Liu	Canceller-Kiyani	Canceller-Lampu	Proposed Canceller
NMSE (dB)	-38.89	-39.96	-41.31	-43.82
ACEPR-Left (dBc)	-53.26	-57.01	-59.39	-66.18
ACEPR-Right (dBc)	-52.20	-55.71	-58.97	-66.28
FLOPs/sample	796	809	3309	651

For the 100 MHz contiguous 5G-NR signal, the average RF output power of the power amplifier (PA) is 32.71 dBm. The circulator used in the system provides an isolation of 40 dB, resulting in a transmitter leakage level of -7.29 dBm at the receiver input prior to the LNA.

Fig. 5 illustrates the power spectral density of the transmitter leakage signal both without cancellation method and after applying different nonlinear cancellation methods for contiguous 5G-NR signal. The proposed cancellation method achieves the highest suppression level and yields the lowest residual leakage compared to existing nonlinear cancellers.

Table 2 summarizes the power levels and cancellation performance for the transmitter leakage signal at the receiver under different cancellation configurations. Without a canceller method, the leakage power level at the receiver remains at -7.29 dBm with 40 dB circulator isolation. As shown in Table 2, the proposed canceller achieves significantly lower residual power and a higher cancellation margin relative to other state-of-the-art approaches for the 100 MHz contiguous 5G-NR waveform. The proposed canceller demonstrates approximately 7 dBm lower power compared to the other

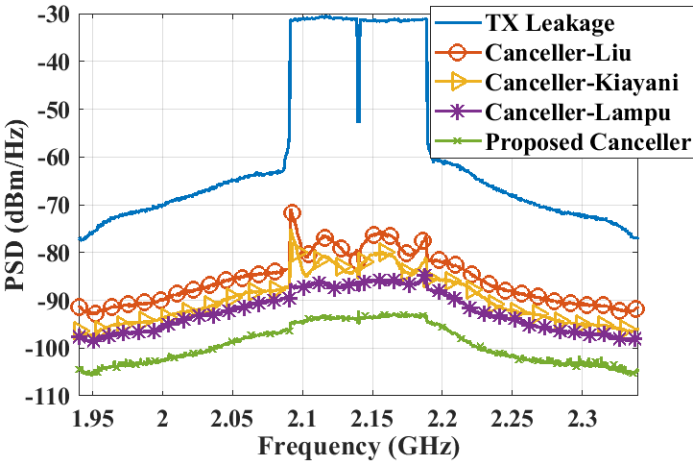


Fig. 5 Power Spectrum Density (PSD) of Transmitter's leakage for contiguous 5G-NR signal.

cancellation techniques. It also provides about 7 dB higher cancellation efficiency than the existing cancellers and achieves more than 100 dB SI cancellation.

Table 2 Cancellation Results of Nonlinear Canceller Models for 100 MHz Contiguous 5G-NR Signal

	Tx Leakage w/o canceller	Canceller-Liu	Canceller-Kiyani	Canceller-Lampu	Proposed Canceller
Power (dBm)	-7.29	-52.40	-56.33	-60.34	-67.54
Cancellation (dB)	40	85.11	89.04	93.05	100.25

Fig. 6 shows the cancellation results of nonlinear cancellation methods with several transmitted powers for 100 MHz Contiguous 5G-NR Signal. The cancellation (dB) is measured for different cancellation methods with varying transmitter (PA) powers at the receiver. The proposed cancellation method provides higher cancelation (dB) than the existing nonlinear cancellers for different transmitted powers.

3.3 Measurement Result for Non-Contigous Signal in Flexible-Duplex

Table 3 presents the modeling results of several advanced nonlinear cancellation approaches for 100 MHz non-contiguous 5G-NR waveform. For performance evaluation, the proposed NN model is configured with M=4, h_1=9, and h_2=8. Experimental results demonstrate that the proposed cancellation model achieves a lower NMSE with less computational complexity compared to the existing nonlinear cancellers for the 100 MHz non-contiguous 5G-NR signal. Moreover, the proposed canceller provides an enhancement greater than 8 dBc in both ACEPR-Left and ACEPR-Right, demonstrating better modeling performance relative to other referenced approaches.

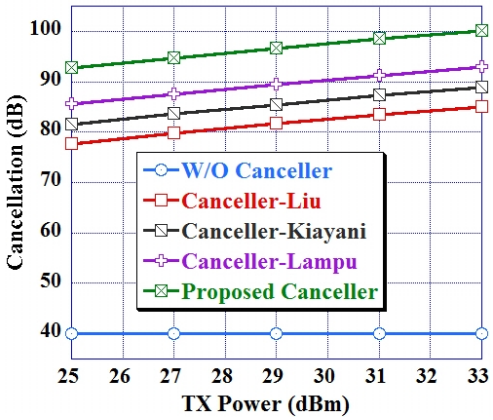


Fig. 6 Cancellation results of Nonlinear Canceller Models with several Tx Powers for 100 MHz Contiguous 5G-NR Signal.

Table 3 Modeling Results of Nonlinear Canceller Models for 100 MHz Non-Contiguous 5G-NR Signal

Metric	Canceller			
	Canceller-Liu	Canceller-Kiyani	Canceller-Lampu	Proposed Canceller
NMSE (dB)	-40.43	-41.82	-42.57	-44.49
ACEPR Left (dBc)	-57.32	-61.04	-63.22	-71.60
ACEPR Right (dBc)	-57.21	-60.31	-64.00	-72.12
FLOPs/sample	796	809	3309	651

For the 100 MHz non-contiguous 5G-NR signal, the average RF output power of the power amplifier (PA) is 32.33 dBm. The circulator used in the system provides an isolation of 40 dB, resulting in a transmitter leakage level of -7.67 dBm at the receiver input prior to the LNA.

Fig. 7 illustrates the power spectral density of the transmitter leakage signal both without cancellation method and after applying different nonlinear cancellation methods for non-contiguous 5G-NR signal. The proposed cancellation method achieves the highest suppression level and yields the lowest residual leakage compared to existing nonlinear cancellers.

Table 4 summarizes the power levels and cancellation performance for the transmitter leakage signal at the reciever under different cancellation configurations. Without a canceller method, the leakage power level at the receiver remains at -7.67 dBm with 40 dB circulator isolation. As shown in Table 4, the proposed canceller achieves significantly lower residual power and a higher cancellation margin relative to other state-of-the-art approaches for the 100 MHz non-contiguous 5G-NR waveform. The proposed canceller demonstrates approximately 9 dBm lower power compared to the

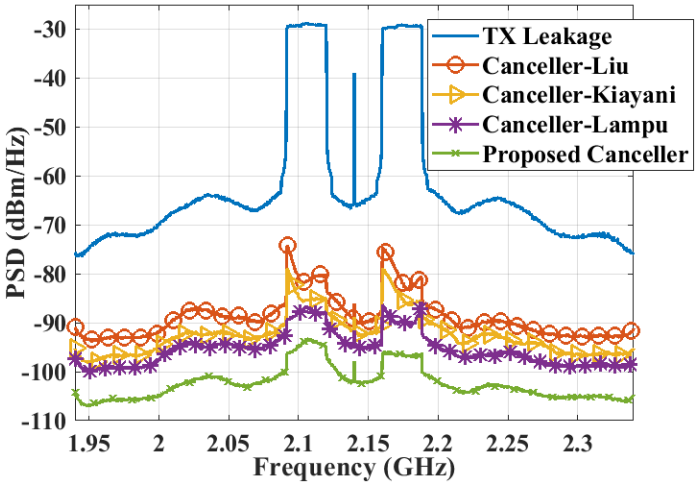


Fig. 7 Power Spectrum Density (PSD) of Transmitter's leakage for non-contiguous 5G-NR signal.

Table 4 Cancellation Results of Nonlinear Canceller Models for 100 MHz Non-Contiguous 5G-NR Signal

	Tx Leakage w/o canceller	Canceller-Liu	Canceller-Kiyani	Canceller-Lampu	Proposed Canceller
Power (dBm)	-7.67	-54.01	-57.98	-62.21	-70.85
Cancellation (dB)	40	86.34	90.31	94.54	103.18

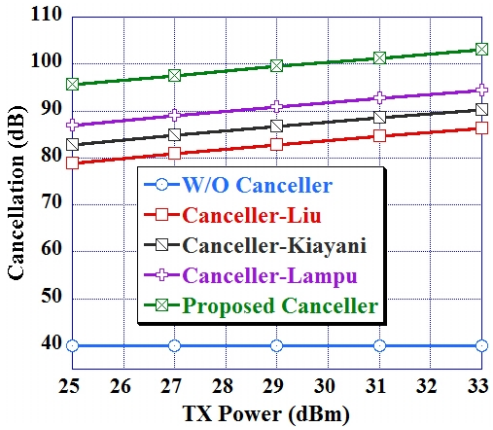


Fig. 8 Cancellation results of Nonlinear Canceller Models with several Tx Powers for 100 MHz Non-Contiguous 5G-NR Signal.

other cancellation techniques. It also provides about 9 dB higher cancellation efficiency than the existing cancellers and achieves more than 103 dB SI cancellation.

Fig. 8 shows the cancellation results of nonlinear cancellation methods with several transmitted powers for 100 MHz non-contiguous 5G-NR Signal. The cancellation (dB) is measured for different cancellation methods with varying transmitter (PA) powers at the receiver. The proposed cancellation method provides higher cancelation (dB) than the existing nonlinear cancellers for different transmitted powers.

4 Conclusion

A NN-based self-interference cancellation architecture is proposed for emerging 6G flexible-duplex systems. By accurately modeling nonlinear behavior in GaN HEMT PAs, incorporating long-term memory effects, and accounting for low-noise amplifier distortion, the proposed solution enables both analog-domain and digital-domain cancellation. Measurement results for 100 MHz contiguous and non-contiguous NR waveforms demonstrate that the proposed canceller significantly outperforms existing nonlinear cancellation schemes. The proposed model achieves better NMSE and ACEPR values with less computational complexity while consistently reducing transmitter leakage power by 7–9 dBm as compared to other nonlinear cancellers in contiguous and non-contiguous configurations. Furthermore, the proposed architecture achieves more than 100 dB suppression of SI in contiguous scenarios and over 103 dB suppression of SI in non-contiguous scenarios, confirming its effectiveness in highly dynamic duplexing environments. These results validate the proposed method as a strong candidate for self-interference mitigation in future flexible-duplex and full-duplex communication systems, supporting the high spectral efficiency demands expected in 6G Communication.

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Data availability

The datasets generated and analysed during this study are included in this article.

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